

AUTOMOTIVE ANALYTICS SOLUTIONS

for the Indian Market

Executive Summary

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June 2026

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Executive Summary

India's automotive industry is one of the world's fastest-growing economic sectors — ranked third globally by production volume, contributing approximately 7–8% of GDP and generating around US\$250 billion in annual output (2025). It is a critical export engine, accounting for roughly 8% of the country's total exports.

Yet the sector is facing increasing pressure. The shift to electric vehicles, the rise of software-defined vehicles, and the complexity of multi-tier global supply chains are creating new challenges. Warranty costs — currently running at 2–4% of revenue — are climbing. Supply disruptions are an ever-present risk. And despite vast amounts of data being generated across OEMs, suppliers, and dealer networks, most of it remains siloed and underused.

Our analytics consultancy addresses these challenges with three integrated, data-driven solutions tailored specifically for Indian automakers and their partners:

- Warranty Analytics — detect quality issues early, reduce claim costs, and prevent recalls.
- Supply-Chain Risk Prediction — monitor supplier health and logistics in real time, and act before disruptions occur.
- Dealer & Field Service Intelligence — turn dealer data into actionable insights that boost sales performance and customer satisfaction.

Together, these solutions have demonstrated measurable returns: leading OEMs using proactive quality analytics report 5–10% reductions in warranty expense; AI-driven supply-chain planning can cut logistics costs by up to 15% and reduce excess inventory by 35%; and structured dealer analytics consistently lifts KPI achievement rates by 10–20%.

This document presents our full offering — covering the Indian market context, detailed service descriptions with KPIs, data requirements, technical approach, implementation roadmap, pricing models, case studies, FAQs, and contact information.

US\$250B	5–10%	15%	~\$20M
India Auto Market (2025)	Warranty Cost Reduction	Logistics Cost Savings	Saved by a Single OEM

Company Overview

We are an independent analytics consultancy specialising in the automotive sector. Our team brings together automotive engineers, supply-chain specialists, data scientists, and software engineers — all focused on solving the industry's most pressing data challenges.

We deliver complete, end-to-end solutions: from data integration and machine learning model development, through to interactive dashboards, automated alerts, and ongoing operational support.

Our approach is practical and client-focused. We work closely with OEMs, Tier-1 suppliers, and dealer groups to tailor each solution to the client's existing data, systems, and business priorities. Engagements are flexible — available as fixed-scope projects or ongoing managed analytics services.

Data security and intellectual property ownership are central to how we operate:

- All client data is kept strictly confidential under a signed NDA.
- All models, dashboards, and outputs developed on your data become your property.
- We apply industry-standard security throughout — encryption in transit and at rest, role-based access controls, and full audit logging.

Market Context — India Automotive

India's automotive sector is a proven growth engine. With production exceeding 2.9 million vehicles in January 2026 alone, it ranks third globally behind China and the United States. Two-wheelers account for approximately 77% of total volume, while passenger vehicles represent around 16%. The sector contributes 7–8% of GDP and a comparable share of national exports.

Growth momentum is strong: government incentives such as PLI schemes and the auto mission plan, combined with major manufacturer investments — including Maruti's Gujarat plant, Volkswagen's Pune facility, and new EV-related projects — are accelerating expansion. EV sales have been doubling year-on-year, and over 250 new models are expected to enter the market in the coming years.

Key Market Pressures

Despite this growth, OEMs and suppliers face significant operational challenges:

- **Warranty Costs:** Global warranty expenses have nearly doubled since 2012, reaching approximately \$58 billion in 2024 (around 2.2% of global auto sales). For some Indian OEMs, this figure exceeds 4% of revenue — a direct drain on profitability.
- **Supply-Chain Complexity:** India's logistics costs run approximately 30% higher as a percentage of sales compared to China, due to infrastructure constraints and inflationary pressures. Multi-tier supplier networks are difficult to monitor and slow to respond to disruptions.
- **Demand Volatility:** EV adoption curves, GST policy changes, and post-pandemic recovery patterns all create forecasting uncertainty that traditional systems struggle to handle.
- **Aftermarket Growth:** As the vehicle parc expands, the service and parts aftermarket is growing rapidly — an opportunity that requires better data to capture.

Why Analytics — and Why Now

The answer to these pressures is not more data — it is better use of the data that already exists. Manufacturers and dealers already collect enormous volumes of information through warranty systems, telematics, DMS platforms, and supply-chain tools. The gap is in turning that data into timely, actionable decisions.

Our solutions are designed to close exactly that gap — giving Indian OEMs and their partners predictive intelligence rather than reactive reporting.

Service Descriptions

1. Warranty Analytics

Warranty claims are one of the most costly and reputationally sensitive areas in automotive operations. A single undetected defect pattern — left unaddressed for weeks — can escalate into a full recall. Our Warranty Analytics service is designed to catch these issues early, enabling quality teams to act before problems reach the field at scale.

What We Do

We consolidate all warranty-related data into a unified analytics environment — including dealer claims, repair orders, parts usage records, vehicle telematics, and quality test results. Using machine learning, we detect emerging defect patterns, predict component failure rates, and identify root causes — often weeks before the issue would surface through conventional reporting.

Key Capabilities

- Time-series anomaly detection on claim volumes and trends
- Clustering of failure symptoms to surface hidden defect patterns
- Correlation analysis linking defects to specific suppliers, plants, or production batches
- Predictive models estimating failure probability within a defined window (e.g., next 6 months)
- Early-warning alerts triggered by statistically significant claim spikes

Business Value

McKinsey research shows that OEMs using AI-driven quality management can reduce warranty expenses by 5–10% or more. Root-cause analysis that typically takes 7+ weeks manually can be compressed to days. One OEM that linked battery telematics to warranty claim data identified defects weeks earlier than usual, avoiding approximately \$20 million in costs within six months.

KPIs Tracked

KPI	Description
Defects per 1,000 Vehicles	Industry-standard quality metric at model and component level
MTBF	Mean Time Between Failures — tracks component reliability over time
Warranty Spend per Unit	Total warranty cost normalised per vehicle produced or sold
Claim Cycle Time	Average time from claim submission to resolution

Recall Incidence Rate	Frequency of recall events triggered by warranty patterns
Cost Avoidance	Estimated savings from early detection before escalation

Deliverables

- **Periodic reports summarising failure trends, root-cause findings, and recommended actions.**Reports:
- **BI dashboards (Power BI / QuickSight) showing claim volumes, defect heatmaps, trending issues, and supplier risk indices.**Dashboards:
- **Automated email/SMS alerts for significant claim spikes or quality metric drops.**Alerts:
- **Trained forecasting and classification models integrated via API into your PLM or ERP systems.**ML Models:

2. Supply-Chain Risk Prediction

A single supplier failure or port delay can halt an entire production line within days. In today's multi-tier supply networks, the challenge is not just visibility — it is anticipating problems before they occur. Our Supply-Chain Risk Prediction service gives planners the intelligence to act early, re-route, and protect production continuity.

What We Do

We ingest data from across the supply chain — supplier profiles, inventory and production plans, shipment tracking feeds, and external signals such as weather, port status, commodity prices, and financial health indicators. AI models assess risk at every node in the network, simulate the downstream impact of potential disruptions, and continuously update risk scores as new information arrives.

Key Capabilities

- Graph-based analysis of multi-tier supplier networks to map dependencies and vulnerability points
- Real-time risk scoring for suppliers, components, and logistics routes
- Disruption simulation — modelling the production impact of a supplier failure or material shortage
- Predictive lead-time forecasting for critical parts
- Monitoring of external risk signals: weather events, political developments, port closures, supplier financial stress

Business Value

Industry analysis shows that AI-driven supply-chain management can cut logistics costs by up to 15% and reduce excess inventory by approximately 35%. Toyota's application of AI in routing reduced delivery lead times by 17% and freed significant working capital through a 22% reduction in excess stock. Nissan reported avoiding seven-figure losses over six months by acting on early risk alerts from a supply-chain control tower.

KPIs Tracked

KPI	Description
Supplier Risk Score	Composite score combining on-time delivery, financial health, and external signals
Parts Availability	Percentage of required parts available on hand at any given time
Days of Inventory	Number of days of production that current stock can support
Fill Rate	Percentage of orders fulfilled on time and in full
OTIF	On-Time-In-Full delivery rate across the supplier network
Production Downtime	Hours/days of lost production attributable to supply disruptions

Deliverables

- **Control-tower view showing a live map of shipments, risk-indexed routes, and a ranked list of at-risk suppliers and components.** Dashboard:
- **Customisable early-warning notifications (email/SMS) when supplier or shipment risk exceeds defined thresholds.** Alerts:
- **Weekly risk summaries including new high-risk items, scenario analyses, and recommended responses.** Reports:
- **Predictive engines for lead-time forecasting, demand spike anticipation, and re-order optimisation.** Models:

3. Dealer & Field Service Intelligence

Dealer networks are where the customer relationship — and a significant portion of revenue — is won or lost. Yet most OEMs lack a consistent, real-time view of what is happening across their dealer base. Our Dealer & Field Service Intelligence service turns fragmented dealer data into a single, coherent picture that management can act on.

What We Do

We connect to Dealer Management Systems (DMS) and CRM platforms to aggregate data on sales performance, finance penetration, service bookings, parts turnover, and warranty upsell rates. Analytics identify patterns — such as which dealers are underperforming on conversions, where service capacity is mismatched with demand, or which regions have low extended-warranty uptake. For field service teams, we incorporate visit logs and technician tracking to forecast workload and optimise dispatch.

Key Capabilities

- Consolidated dealer performance dashboard across the entire network
- Identification of underperforming dealers by sales, service, and aftermarket metrics
- Demand forecasting for service appointments by region and vehicle type
- Lead-scoring models estimating conversion likelihood for incoming prospects
- Technician scheduling optimisation based on predicted service demand

Business Value

BI software designed for automotive consolidates data from sales, finance, service, and administration into a single source of truth — enabling the kind of accountability and targeted coaching that drives real performance improvement. Networks that have adopted structured dealer analytics have reported 15% increases in extended-warranty sales within a year and KPI achievement rate improvements of 10–20%.

KPIs Tracked

KPI	Description
Sales per Dealer	Units sold per dealer over a defined period
Showroom Conversion Rate	Percentage of walk-in or lead enquiries converted to a sale
Finance Penetration	Share of vehicle sales where the dealer arranges financing
Aftermarket Revenue per Vehicle	Parts and service revenue generated per vehicle sold
Warranty Penetration %	Share of vehicles sold with an extended warranty

MTTR	Mean Time to Repair — service efficiency metric
First-Time Fix Rate	Percentage of service visits resolved without a repeat visit
CSI Score	Customer Satisfaction Index — standardised quality of service metric

Deliverables

- **Network-wide dealer scorecards with location-level drill-down and KPI red/amber/green flags.**Dashboards:
- **Automatic notifications for low finance penetration, rising repeat-visit rates, or significant KPI drops.**Alerts:
- **Monthly dealer scorecards and field service performance reports.**Reports:
- **Service demand forecasting and lead-conversion probability models.**Models:

Data Requirements & Sources

Each service draws on a combination of internal and external data. We work with clients during Phase 1 to catalogue all available data assets, assess quality and completeness, and define a practical integration plan. The table below summarises the main categories:

Data Source	Examples	Used In
Internal (OEM / Dealer)	Warranty claims, repair orders, production logs, dealer management system (DMS) records, service logs, telematics & IoT data, CRM and call-centre logs, procurement data, quality test results, technician schedules	All three services
OEM Filings & Industry	Recall/defect bulletins, ARAI & NHTSA recall databases, SIAM production statistics, Tier-1 supplier financial reports, compliance documents	Warranty, Supply-Chain
Public / Government	VAHAN vehicle registration data, Ministry of Commerce import/export stats, traffic and road condition feeds, weather and GIS data	Supply-Chain, Dealer
Third-Party / Partners	Market research and industry forecasts, logistics & GPS tracking feeds, commodity price data (steel, semiconductors), social media sentiment and product reviews	All three services

In practice, we extract and normalise data through ETL pipelines, typically storing raw data in secure cloud environments (AWS S3, Azure Blob Storage) or connecting to on-premise databases where required. Data quality and governance processes are embedded from day one.

Technical Approach

We deploy a modern, cloud-based (or hybrid) data and ML architecture designed for reliability, scalability, and security. The platform is structured around five core layers:

1. Data Ingestion & Lake

Data is collected through secure connectors — database queries, REST APIs, and file transfers — and landed in a centralised data lake with role-based access controls, encryption at rest and in transit, and VPC-level network isolation.

2. Data Pipeline & Transformation

ETL/ELT tools (AWS Glue, Azure Data Factory, or equivalent open-source pipelines) clean, standardise, and structure incoming data. Warranty defect codes are harmonised, supplier networks are mapped, and dealer identifiers are unified across systems. A metadata catalogue supports data governance and lineage tracking.

3. Analytics & Machine Learning

Data scientists build and maintain ML models using Python-based frameworks (scikit-learn, TensorFlow, PyTorch) or managed ML platforms (AWS SageMaker, Azure ML). Models include time-series forecasters (LSTM, ARIMA), anomaly detectors, classifiers, and graph-based network analysis tools. All models are built to be explainable — producing feature importance scores and root-cause flags, not just predictions.

4. Knowledge Layer (RAG)

A Retrieval-Augmented Generation (RAG) layer is embedded for knowledge management — allowing, for example, a warranty analytics agent to query technical manuals, recall notices, and repair guides to generate context-rich explanations. This supports natural-language Q&A interfaces and makes the platform accessible to non-technical users.

5. Visualisation, Alerts & Security

Model outputs feed into BI dashboards (Power BI, Tableau, AWS QuickSight) with role-based views — summary KPIs for executives, analytical detail for data teams. Alerts are delivered via email or messaging platforms (AWS SNS or equivalent). The platform is containerised for flexible scaling and can be deployed on-premise, in the cloud, or in a hybrid configuration.

Data Flow (Example): Raw dealer data → Secure Data Lake → ETL pipeline (cleanse & aggregate) → Data Warehouse → ML models (forecast demand, score anomalies) → BI dashboards → Role-based alerts and reports

Implementation Roadmap

We recommend a phased approach over 9–12 months, structured to deliver value progressively rather than requiring a long wait for results. Clients begin seeing working dashboards and alerts within the first three months.

Phase	Timeline	Key Activities	Deliverables
Phase 1 Discovery	Months 1–3	Data audit & requirements gathering, system access setup, baseline dashboard prototype	Data integration plan, PoC dashboard, requirements sign-off
Phase 2 Development	Months 4–7	ML model development, ETL pipeline build, dashboard finalisation, pilot testing with end users	Trained models, tested dashboards, pilot results report
Phase 3 Rollout	Months 8–12	Full deployment to all stakeholders, user training, model monitoring setup, optimisation based on feedback	Production-ready platform, training materials, SLA-backed support

Each phase is designed to stand on its own: Phase 1 produces a working proof-of-concept; Phase 2 delivers production-grade models; Phase 3 achieves full operational deployment with monitoring and optimisation in place.

Pricing & Engagement Models

We offer flexible engagement structures to suit different client needs — from one-time project builds to long-term managed service partnerships. All pricing is custom-quoted after an initial scoping session; the table below describes the available models.

Engagement Model	Description	Best Suited For
Fixed-Price Project	Defined scope delivered for a lump-sum fee. Covers one or two phases.	Well-scoped requirements, first-time engagements
Time & Materials	Iterative work billed by daily or weekly effort rates.	Exploratory / R&D-intensive projects
Managed Service / Subscription	Monthly retainer covering hosting, maintenance, monitoring, and periodic reports.	Ongoing analytics with continuous updates
Revenue-Share / Success Fee	Co-development with fee tied to agreed KPI improvements.	Performance-driven partnerships

Indicatively, analytics engagements typically range from a few lakhs (for small proof-of-concepts) to several crores of rupees for full enterprise-scale deployments. Final pricing is confirmed following the discovery phase. We can also provide a phased commercial structure where costs align with each phase of delivery.

Case Studies & ROI Examples

Warranty Analytics — \$20M Saved in Six Months

A leading global OEM integrated vehicle telematics data with its warranty claim database using predictive analytics. By identifying defect patterns weeks earlier than traditional methods allowed, the company was able to intervene before failures reached the field at scale. The result: approximately \$20 million in avoided costs within six months. McKinsey research corroborates this, noting that proactive quality analytics can reduce warranty costs by 5–10%. For an Indian manufacturer spending ₹500 crore per year on warranty, a 5% reduction alone equates to ₹25 crore in annual savings.

Supply-Chain Risk — Nissan Avoids Seven-Figure Losses

Nissan deployed a supply-chain control tower with AI-based risk monitoring — tracking weather events, port status, logistics delays, and supplier financial health in real time. Over a six-month period, the system's early alerts allowed the company to re-route and pre-empt disruptions, avoiding losses that would otherwise have run into the millions. Separately, global benchmarks show that AI-driven supply-chain optimisation can reduce logistics costs by up to 15% and lower inventory by approximately 35%. Toyota's AI-assisted routing programme cut delivery lead times by 17% and reduced excess stock by 22%.

Dealer Intelligence — 15% Increase in Extended-Warranty Sales

A dealer network that implemented structured analytics across its outlets — tracking warranty penetration, finance rates, and service KPIs — reported a 15% increase in extended-warranty sales within one year. The ability to identify and coach underperforming dealers based on objective data (rather than anecdotal reporting) lifted overall KPI achievement rates by 10–20% across the network.

These examples illustrate a consistent pattern: analytics investments in automotive operations typically pay back within 12 months. A conservative model — a 5% reduction in warranty costs combined with a 2% uplift in dealer revenue — is often sufficient for the solution to be self-funding. We can provide a custom ROI projection once we understand your specific data and cost baseline.

Sample Outreach Materials

One-Pager Highlights

The following outline can be used as the basis for a client-facing one-pager or leave-behind document:

- **Transform Your Quality, Supply, and Dealer Operations with AI-Driven Analytics.**Headline:
- **India is the world's third-largest auto market — and rising complexity is creating costly new risks.**Market Context:
- **Rising warranty spend, unpredictable supply disruptions, inconsistent dealer performance.**Pain Points:
- **Warranty Analytics (catch defects early), Supply-Risk AI (prevent production stoppages), Dealer BI (maximise network performance).**Our Three Solutions:
- **\$20M saved by one OEM through early defect detection; 15% logistics cost reduction through AI supply-chain management; 15% increase in warranty sales through dealer analytics.**Proof Points:
- **Pilot offer available — contact us to discuss a limited-scope proof-of-concept.**Next Step:

Presentation Slide Outline

For executive presentations, we recommend the following structure:

- **Title — Company name, solution name, date**Slide 1:
- **India Market Context — size, growth, complexity**Slide 2:
- **Warranty Analytics — the problem, our approach, sample dashboard**Slide 3:
- **Supply-Chain Risk — the problem, solution example, map visual**Slide 4:
- **Dealer Intelligence — the problem, approach, KPI chart**Slide 5:
- **Data & Technology — architecture overview, data sources**Slide 6:
- **Implementation Roadmap — phased Gantt view**Slide 7:
- **Pricing & Engagement Models — summary table**Slide 8:
- **Case Studies & ROI — key results and savings**Slide 9:
- **FAQs**Slide 10:
- **Call to Action & Contact Information**Slide 11:

Frequently Asked Questions

Q: What data do we need to get started? We work with whatever structured data is available. The minimum viable starting point includes historical warranty claims (with dates and defect codes), a supplier and parts catalogue, and dealer sales logs. Richer data — such as telematics, CRM, and procurement records — produces more sophisticated models. Data cleansing and governance are embedded into Phase 1, so incomplete or inconsistent data is not a barrier to starting.

Q: Who owns the models and outputs? You do. All models, dashboards, and code developed using your data are delivered to you and become your intellectual property. In a managed service arrangement, we retain working copies for maintenance purposes, but full IP ownership remains with the client.

Q: What if we have limited or messy data? This is common, and it does not prevent us from proceeding. Where key data sources are absent — such as telematics — we rely more heavily on claim histories and repair logs, and may supplement with industry benchmarks or transfer learning. Phase 1 is specifically designed to assess data readiness and build a plan around what exists.

Q: How quickly will we see results? Working dashboards and early alerts are typically available within 3–4 months (end of Phase 1). Measurable ROI — cost savings or efficiency gains — generally materialises by the end of Phase 2 (months 6–9). Timelines depend on the speed of data access and user adoption within the client organisation.

Q: How is pricing structured? All pricing is custom-quoted following an initial discovery session. Costs are denominated in INR and vary with scope and data volume. Small proof-of-concepts typically start in the low lakhs; full enterprise deployments may run to several crores. We can structure payment across phases to align with delivery milestones.

Contact & Next Steps

We welcome the opportunity to discuss how our analytics solutions can address your specific challenges. The typical engagement process is straightforward:

- **A 30–60 minute conversation to understand your pain points, data landscape, and priorities.**Step 1 — Initial Call (no cost):
- **A structured scoping exercise to define requirements, assess data readiness, and agree on success metrics.**Step 2 — Discovery Workshop (1–2 weeks):
- **A detailed commercial and technical proposal, with phased pricing and a clear roadmap.**Step 3 — Proposal:
- **For clients who prefer to start small, we offer a limited-scope proof-of-concept for one service at a reduced fee — a low-risk way to validate the approach before full commitment.**Step 4 — Pilot Option:

Contact	Details
Email	contact@our-consultancy.ai
Phone	+91-XXXXXXXXXX
Engagement	OEMs, Tier-1 Suppliers, Dealer Groups
Confidentiality	All discussions are covered by NDA on request

Sources: This document draws on publicly available industry and government reports, analytics solution guides, and validated case studies from McKinsey, Toyota, Nissan, and others. All market statistics and ROI figures cited are referenced from reputable third-party sources.